Lesion-Anchored Prototypes: Reducing Artifact Shortcuts in Interpretable Melanoma Classification

Yousef Azizi Movahed
Dept. of Computer Science, School of Mathematics, Statistics, and Computer Science, College of Science
University of Tehran, Tehran, Iran
azizi.movahed@ut.ac.ir

Fatemeh Ziaeetabar*
Dept. of Computer Science, School of Mathematics, Statistics, and Computer Science, College of Science
University of Tehran, Tehran, Iran
fziaeetabar@ut.ac.ir

***Abstract*—Melanoma classifiers pick up artifacts such as dark corners, rulers and ink, and a prototype network's explanation ("this part looks like that prototype") may point at the artifact. On ISIC 2019, dark corners and bright structures remain associated with melanoma after stratifying by source. Existing remedies supervise accuracy or activations, but in dermoscopy none controls *which* training patches prototypes may become. We propose lesion-anchored prototype learning with two components. A lesion-attraction objective acts on each prototype's normalized spatial attention, and the projection step admits only cells that are at least half lesion and nearly free of detected artifacts. Every prototype thus comes from a lesion-dominated patch by construction, although its receptive field extends beyond it. The masks need no new annotation (Dice 0.94 on held-out expert masks); inference needs none. In a same-backbone, leakage-audited comparison with lesion-grouped statistics, the model raises the AUC of ProtoPNet from 0.825 to 0.865 (0.859 without an auxiliary artifact-suppression term). When artifacts are inserted around bit-identical lesion pixels, lesion attraction roughly halves ProtoPNet's decision-flip rate (dark corners: 10.9%→5.2%). Both gains hold across seven runs per model ($p \leq 0.03$), and ProtoPNet with the same projection stays less robust. The strongest mask-supervised ProtoPNet reaches similar accuracy (0.867) and, across runs, robustness, but put all prototypes on lesion-dominated cells in one of seven runs; our projection does so in every run at no accuracy cost. Adding that mask loss weakens the real dark-corner shortcut further, though alone it generalizes better to Derm7pt. Code and logs: https://github.com/YousefAMovahed/lesion-anchored-prototypes.**

***Index Terms*—explainable AI, prototypical part networks, shortcut learning, dermoscopy artifacts, melanoma, counterfactual evaluation**

I. Introduction

Convolutional networks can classify skin lesions as accurately as dermatologists [1], yet they also learn *shortcuts* [2]: features that correlate with the label in the training data but are unrelated to the disease. In dermoscopy, surgical skin markings raise the melanoma probability of a deep classifier [3], models perform far above chance even when the lesion is removed [4], [5], and dark corners, rulers, gel bubbles and ink are common and unevenly spread across sites [6]–[8].

Rudin [9] argues that clinical use calls for models that are interpretable by design. The prototypical part network (ProtoPNet) [10] compares image patches with learned prototypes, each projected onto a real training patch, so a decision reads "this part looks like that part". Yet nothing prevents a prototype from being a piece of a ruler or a black corner. The shortcut then survives and is even presented as a reasonable explanation [11]; such prototypes have been seen in dermoscopy [12], [13].

Research gap. Three questions remain open (Table I). First, debiasing methods for dermoscopy [7], [14], [15] improve robustness to artifacts, but their models are black boxes. Second, in prototype networks, mask supervision penalizes activation outside annotated regions [12], [16], and human feedback removes or switches off unwanted prototypes [13], [17]. None of these methods constrains the *push* step, which decides which training patch a prototype finally *is*, and feedback helps only once a user has spotted the problem. Third, artifact insertion has been used to test the *decisions* of classifiers [13], [15], [18], but, to our knowledge, not whether the *explanations* of a prototype network stay on the lesion when artifacts are added or removed.

Contributions. We address these gaps:

- **Lesion attraction on the normalized prototype attention.** An objective on each prototype's spatial attention pulls it toward the lesion without shrinking activations. It needs no new annotation and drives the accuracy and robustness gains: without it, the test AUC drops from 0.865 to 0.830 ($p < 0.001$) and flip rates under inserted dark corners, rulers and ink roughly double. The main model gains 4.0 AUC points over ProtoPNet (3.4 with the two core components alone), also across seven runs per model, and ProtoPNet with the same projection still flips 1.8 times as often under dark corners. Across runs, accuracy and robustness are on par with the mask-supervised ProtoPNet of the closest prior work [12], whose artifact share varies from 0.5% to 24.6% between runs (ours 3.0–8.9%).
- **Lesion- and artifact-constrained projection.** The push step admits only latent cells whose input patch is at least half lesion and at most 5% detected artifact, so every prototype comes from one *by construction*. Mask losses give no such guarantee (all 20 prototypes on such cells in one of 19 runs). In 12 networks of three training objectives projected both ways, the step changes neither the AUC nor, consistently, flip rates or explanation maps; thresholds of 0.3–0.9 behave alike. It thus fixes which patch an

*Corresponding author: Fatemeh Ziaeetabar (fziaeetabar@ut.ac.ir).

explanation shows and barely affects decisions. Unlike a concurrent foreground-restricted projection for natural images [19], it excludes artifacts and is combined with lesion attraction and tested against shortcuts.

- **A counterfactual test of decisions *and* explanations.** We insert four artifact types and remove real hair and ruler marks while keeping every lesion pixel bit-identical. With an inpainting-only null control and a complementary test on unedited images, we measure how far both the decision and the prototype explanation move.
- **A leakage-controlled, mask-audited evaluation.** We group near-duplicates in addition to lesion identifiers and validate every mask source and the artifact detectors. Eight configurations, four further variants of the mask-loss baseline and its combination with our model use the same backbone and three seeds (PPN, the Correia formulation and Ours: seven); we compare them with lesion-grouped statistics, across training runs and on three external sets.

The main model also keeps an auxiliary artifact-suppression term; it brought no significant gain on its own, so we report it as a largely negative result rather than a contribution.

II. Related Work

Artifacts and shortcuts in dermoscopy. Bissoto *et al.* annotated seven artifact types, found that simple debiasing does not remove their bias [5] and built artifact-based domain-generalization benchmarks [6]. Removing dark corners barely changes accuracy but moves activations onto the lesion [8], and pasted colored patches make a CNN call malignant lesions benign [18]. EPVT [7] learns domain prompts for artifact groups, Bevan and Atapour-Abarghouei [14] unlearn artifact information adversarially, and Reveal-to-Revise (R2R) [15] finds band-aids, rulers and skin markers semi-automatically, corrects the model with RRR, CDEP or ClArC and tests it with pasted artifacts. All these models are black boxes.

Prototype networks and explanation supervision. The similarity a prototype network suggests can be misleading [11], so explanations should be tested instead of taken at face value [20]. RRR [21] penalizes input gradients inside annotated irrelevant regions, and CDEP [22] penalizes contextual-decomposition importance, demonstrated on ISIC images with colored patches. IAIA-BL [16] adds a fine-annotation loss on prototype activation outside regions marked by radiologists. Closest to us, Correia *et al.* [12] apply this loss to melanoma-versus-nevus classification with lesion masks from a segmentation network; our PPN-FA baselines follow it, in its original and a weaker formulation (Section VI-G). ProtoPDebug [17] and the medical study of PIP-Net [13] let a user mark confounded prototypes, which are then forgotten or switched off. Concurrently, Lin and Wright [19] study architectural prototype diversity on birds and faces, with a hinge loss on the background share of each prototype's activation and, in an appendix, a projection restricted to patches with more than 50% foreground. We instead target shortcut artifacts, exclude them in the projection, need no new mask annotation and test with counterfactuals.

TABLE I

Positioning against related work. Annot.: human input; Interp.: interpretable by design; Where: constrains where the model looks; Prov.: constrains which training patches prototypes become; CF: artifact interventions testing decisions (D) and explanations (E). Lin and Wright [19] study prototype diversity on birds and faces, not artifacts (†appendix variant).

Method	Annot.	Interp.	Where	Prov.	CF
EPVT [7], Bevan [14]	image labels	–	–	–	–
R2R [15]	semi-auto.	–	✓	–	D, E
RRR [21], CDEP [22]	pixel masks	–	✓	–	–
ProtoPNet [10]	none	✓	–	–	–
IAIA-BL [16]	expert masks	✓	✓	–	–
Correia *et al.* [12]	auto. masks	✓	✓	–	–
ProtoPDebug [17]	user	✓	✓	post hoc	–
PIP-Net (med.) [13]	user	✓	–	post hoc	D
Lin, Wright [19]	object masks	✓	✓	foreground†	–
Ours	auto. masks	✓	✓	**les., no art.**	**D, E**

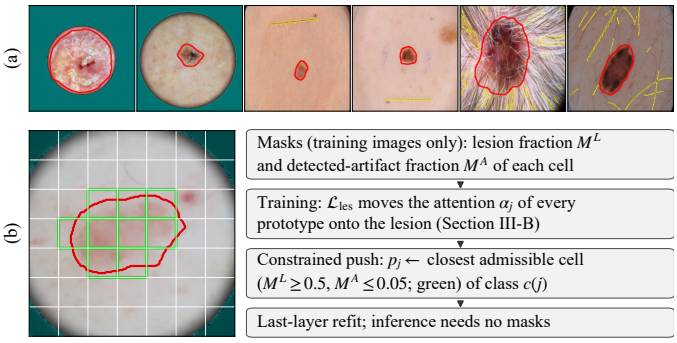


Fig. 1. (a) Automatic masks on ISIC 2019 images (red: lesion; cyan: dark corners; yellow: rulers, third and fourth, and hair, last two); artifact masks never overlap the lesion. (b) Method overview: the 7×7 cells of a test image that the constrained projection would admit (green), and the pipeline.

III. Method

A. Prototypical part network

Our network uses a DenseNet-121 backbone [23] (Fig. 1(b)). Two 1×1 convolutions with a final sigmoid give a latent map $z \in [0,1]^{H \times W \times D}$, $H = W = 7$, $D = 128$. There are $m = 20$ prototypes $p_j \in \mathbb{R}^D$, ten per class; $c(j)$ is the class of prototype j. Each cell has distance $d_j(h,w) = \|z_{hw} - p_j\|_2^2$ and similarity $s_j(h,w) = \log \frac{d_j(h,w)+1}{d_j(h,w)+\epsilon}$ with $\epsilon = 10^{-4}$. A prototype is as active as its best-matching cell, $a_j = \max_{h,w} s_j(h,w)$, and the logits are $\mathbf{W}\boldsymbol{a}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_c \mathcal{L}_{\text{clst}} - \lambda_s \mathcal{L}_{\text{sep}} + \lambda_l \mathcal{L}_{\text{les}} + \lambda_a \mathcal{L}_{\text{art}}, \quad (1)$$

where $\mathcal{L}_{\text{CE}}$ is a class-weighted cross-entropy and the cluster and separation costs measure the distance to the closest prototype of the own and of the other class [10] ($\lambda_c = 0.8$, $\lambda_s = 0.08$ as originally). Our model combines its two core components, lesion attraction $\mathcal{L}_{\text{les}}$ ($\lambda_l = 1$) and the constrained projection (Section III-C), with an auxiliary artifact-suppression term $\mathcal{L}_{\text{art}}$ ($\lambda_a = 1$; Section VI-H), all using the masks of Section IV.

B. Lesion attraction (core component)

Masks are augmented geometrically like the image, and average pooling to the 7×7 grid gives the lesion and artifact fractions of each cell, M^L and M^A. Where a prototype looks

is given by its spatial attention $\alpha_j = \mathrm{softmax}_{h,w}(s_j/\tau)$ with $\tau = 0.5$, a soft version of the max-pooling location behind a_j. Each prototype is weighted by $\omega_j = a_j / \sum_k a_k$, treated as a constant, so that the prototypes the model uses count most. With $L_{ij} = \sum_{h,w} \alpha_{ij} M_i^L$, the attention mass of prototype j on the lesion of image i,

$$\mathcal{L}_{\mathrm{les}} = -\frac{1}{|\mathcal{B}_L|} \sum_{i \in \mathcal{B}_L} \sum_{j:\, c(j)=y_i} \tilde{\omega}_{ij} \log\left(L_{ij} + \delta\right), \qquad (2)$$

where $\tilde{\omega}$ renormalizes ω over the prototypes of the true class y_i, $\delta = 10^{-4}$, and $\mathcal{B}_L$ holds the batch images whose crop still shows lesion. The logarithm penalizes prototypes that almost ignore the lesion far more than those already on it. Since α_j sums to one, a larger lesion mass leaves less for everything else, artifacts included. The IAIA-BL fine-annotation loss [12], [16] penalizes activation magnitude outside a mask, whereas $\mathcal{L}_{\mathrm{les}}$ acts on the normalized attention and leaves the activation scale to the other losses. Unlike the concurrent hinge on the background activation share [19], it rewards the lesion mass logarithmically and weights each prototype by its use.

C. Lesion- and artifact-constrained projection (core component)

In the "push" step, ProtoPNet replaces each prototype with the closest latent patch from a training image of its class. This patch may lie on a ruler, a black corner or healthy skin even when the attention was well placed during training. We therefore admit only cells with $M^L \geq 0.5$ and $M^A \leq 0.05$ (Fig. 1(b)). By construction, every prototype is then the latent vector of a cell whose 32×32 input patch is at least half lesion and at most 5% detected artifact, so the "that part" of each explanation is a lesion-dominated patch. The constraint fixes where a prototype comes from, not every pixel that shapes its latent vector, since a cell also sees its surroundings (Section VI-C). After the push, only $\mathbf{W}$ is trained, with backbone, add-on layers and batch-normalization statistics fixed, so each prototype stays identical to its source patch. Five last-layer epochs did not solve this convex problem for *any* prototype model, so we re-optimize $\mathbf{W}$ to convergence on fixed training activations (weighted cross-entropy plus an ℓ_1 penalty of 10^{-4} on negative-class connections [10]; 3,000 full-batch Adam steps, cosine-annealed learning rate from 0.02), identically for every prototype model and without validation data; we also report results without it.

IV. Masks: Provenance, Reliability and Leakage Control

A. Provenance and validation of the lesion masks

Each image is tagged with the source of its mask. For the 10,015 HAM10000 images (39.5% of ISIC 2019) we use the public expert segmentations of HAM10000 [24] unchanged. The 15,316 BCN20000 [25] and MSK [26] images have no public masks, so their masks are predicted: we train a DeepLabV3 [27] network with an ImageNet-initialized MobileNetV3-Large [28] encoder at 256×256 (12 epochs, AdamW, one-cycle learning rate up to 5×10^{-4}, binary cross-entropy plus soft Dice, random flips and 90° rotations) *only* on the 8,163 HAM10000 images of our training split, which was fixed beforehand. The prediction is averaged with that of the flipped image, thresholded at 0.5, reduced to the largest connected component and hole-filled. Masks covering less than 0.5% of the image (19 images, 0.08%) are marked invalid and excluded from mask-based losses and metrics. On the 1,852 HAM10000 images of the validation and test splits, which the segmentation network never saw, predicted masks reach a mean Dice of 0.9445 against the expert masks (median 0.9672, mean Jaccard 0.9015), and on the 200 expert-segmented PH2 images [29], from another clinic and device, 0.918 without adaptation. Section VI-I tests the dependence on mask quality.

B. Artifact masks

Artifact masks come from fixed, label-free image operations at 256×256 and always exclude the dilated lesion. *Dark corners* (DC) are dark regions (HSV value < 50) touching the border and at least 8 px thick on average. *Thin dark structures* (TS: hair, ruler ticks) are found with a 13×13 black-hat filter as in DullRazor [30] plus an elongation test, *bright structures* (SP: specular highlights, gel-bubble rims, also light hairs) with an 11×11 white top-hat filter, and *ink or colored patches* (INK) as strongly saturated blue-to-violet pixels (Fig. 1(a)). An image is flagged once a mask covers more than 1%, 0.3%, 0.05% or 0.1% of it. Compared with the public manual labels of Bissoto *et al.* [5], available for 2,593 of our MSK images, the flags are precise but conservative (precision/recall: DC vs. dark corner 0.92/0.10, TS vs. hair or ruler 0.96/0.62, SP vs. gel bubble or border 0.86/0.35, INK vs. ink or patches 0.74/0.25); a visual check of 24 random DC and 24 random TS flags agrees (24 and 20 correct). Many artifacts are missed, above all faint dark corners (37% manual vs. 4% detected prevalence). "Without artifacts" therefore means "without *detected* artifacts": an undetected artifact can enter a prototype cell, artifact shares count detected artifacts only, and the reference group of Section VI-E contains undetected ones, which pulls A toward 0.5 for every model. In the 2,092 training images with manual image-level labels, only 1.9% of the admissible cells lie on the outer ring (where dark corners lie) of an image with an undetected dark corner. For hair or rulers, gel and ink, which the labels do not localize, only loose upper bounds are possible (40%, 43%, 8%). None of the 5 prototypes of Ours from these images is such a cell.

C. How masks are used, and why they cannot leak

Masks are used only in the training losses, the constrained projection, the attribution metrics and to place counterfactual edits. The classifier never receives a mask, so test results do not depend on test masks. The split was computed first, from identifiers and image content alone, before any model (including the segmentation network) was trained. Grouping by lesion identifier and near-duplicate pairs (Section V-A) can only remove leakage; an audit found no group, lesion identifier or image shared between sets, and all 342 near-identical pairs

TABLE II
THE ISIC 2019 SPLIT AND THE PREVALENCE OF DETECTED ARTIFACTS. LESION MASKS: EXPERT FOR HAM10000, PREDICTED FOR BCN20000 AND MSK (VALID FOR 99.91, 99.96 AND 100% OF THE TRAINING, VALIDATION AND TEST IMAGES). EXTERNAL SETS: DERM7PT, 1,011 CASES WITH A DERMOSCOPIC AND A CLINICAL IMAGE (252 MELANOMAS); PAD-UFES-20, 2,298 IMAGES (52 MELANOMAS); PH2, 200 IMAGES (SEGMENTATION ONLY).

	Train	Val.	Test	All
Images (% mel.)	20,552 (17.8)	2,459 (19.3)	2,320 (16.8)	25,331 (17.9)
Diagnosis				
Melanoma	3,657	475	390	4,522
Nevus	10,523	1,208	1,144	12,875
Basal cell carc.	2,671	322	330	3,323
Actinic keratosis	693	80	94	867
Benign keratosis	2,095	278	251	2,624
Dermatofibroma	193	16	30	239
Vascular lesion	201	27	25	253
Squamous cell carc.	519	53	56	628
Source (images / melanomas)				
BCN20000	10,037/2,299	1,225/310	1,151/248	12,413/2,857
HAM10000	8,163/917	952/103	900/93	10,015/1,113
MSK	2,352/441	282/62	269/49	2,903/552
Artifact prevalence (%)				
Dark corners	25.5	25.7	25.1	–
Thin dark structures	24.2	25.7	23.4	–
Bright structures	13.8	14.4	13.3	–
Ink / patches	1.6	1.9	2.0	–

(thumbnail correlation ≥ 0.97) and other verified near-duplicate pairs lie within one set. Supervision uses training data only: the segmentation network learns from the 8,163 training-split masks, mask-based losses and projection use training images, and the last-layer re-optimization uses training activations. Thresholds and checkpoints are chosen on validation, and the 1,852 expert masks of validation and test serve only for evaluation. Neither the segmentation network nor the artifact detectors see a diagnosis, so the masks cannot carry label information. Derm7pt, PAD-UFES-20 and PH2 are used for evaluation only; removing external images flagged as possible near-duplicates of ISIC training images (8, 6, 7) changes no external AUC by more than 0.004.

V. EXPERIMENTAL SETUP

A. Data and leakage-free split

The task is melanoma versus all other diagnoses on the 25,331 training images of ISIC 2019 (HAM10000 [24], BCN20000 [25], MSK [26]; 4,522 melanomas, 17.9%), resized to 256×256. Unlike common practice, we keep the 2,084 MSK images without a lesion identifier. To avoid leakage, we group images by lesion identifier and by near-duplicate pairs (perceptual-hash distance ≤ 8 and thumbnail correlation ≥ 0.9 for full and central image). The 13,259 groups are split into ten stratified folds, one for testing and one for validation (about 81/10/9% per source), with almost identical composition (Table II). External test sets are Derm7pt [31], with a dermoscopic and a clinical image for each of 1,011 cases (melanomas include *in situ* lesions and metastases), and the smartphone images of PAD-UFES-20 [32]. PH2 has no diagnoses in our copy and only checks the segmentation.

B. Models and training

All configurations share backbone (ImageNet-pretrained DenseNet-121), input size (224×224), data, split and schedule. **CNN**: backbone, global average pooling, linear layer (black box). **PPN-std**: ProtoPNet with standard augmentation (random crop, horizontal flip). **PPN**: ProtoPNet with strong augmentation (random resized crop, scale $[0.4, 1]$; horizontal and vertical flips; $\pm 20\%$ brightness, contrast, saturation), used by all following models. **PPN-FA**: PPN with the IAIA-BL fine-annotation loss [16] on our lesion masks as in Correia *et al.* [12], with their $\lambda = 10^{-3}$ but computed on the 7×7 maps and averaged over the batch, which weakens it (Section VI-G tests the original formulations and larger weights). **Ours**: lesion attraction, constrained projection and the auxiliary $\mathcal{L}_{\mathrm{art}}$ ($\lambda_l = \lambda_a = 1$ in Eq. (1)). **Ablations**: without $\mathcal{L}_{\mathrm{les}}$ (Ours-noLes), without $\mathcal{L}_{\mathrm{art}}$ (Ours-noArt, the two core components only), with the standard projection (Ours-stdPush).

We train with Adam, batch size 64 and mixed precision: 5 epochs with frozen backbone (add-on layers and prototypes at 3×10^{-3}), 15 epochs of all layers (backbone 10^{-4}, weight decay 10^{-3}, learning rate divided by 10 every 5 epochs), then the projection and 5 last-layer epochs. The CNN trains for the same 20 epochs and keeps its best validation checkpoint. Class weights are 0.608 and 2.81. Prototype models are always evaluated after the projection, with the last-layer epoch chosen on validation. Each configuration runs with seeds 0, 1 and 2 (about 30 min per run on an NVIDIA T4); PPN, Correia (Section VI-G) and Ours also have seeds 3–6, with trained prototypes stored before the projection. Images and masks, stored at 256×256, receive the same random crop and flips before bilinear resampling to the 224×224 input; test images are only resized. After pooling to the 7×7 grid, each cell covers a 32×32 input patch, and attribution maps are compared with masks at 224×224. Loss weights and the eight configurations, including the choice of the main model, were fixed before any test evaluation; later variants and seeds changed none of them.

C. Evaluation protocol

Classification. Each run's threshold is set on ISIC validation (Youden index) and reused for every test set. AUC, balanced accuracy (BAcc), sensitivity and specificity are mean $\pm$ std over seeds; tests use seed ensembles (mean probability).

Attribution. For a prototype model, we upsample the similarity maps of the three most activated prototypes to 224×224; their top 5% of pixels (the 95th-percentile region that ProtoPNet visualizes) form the hot region. For the CNN we use the class-activation map, which with a global-average-pooling head equals Grad-CAM [33] up to a positive factor [34]. We report the share of the hot region on the lesion and on artifacts, whether its maximum falls on an artifact (pointing), and the *enrichment*, i.e., artifact share divided by artifact area (above 1: more attention than the area explains).

Counterfactuals. Outside a 4-pixel margin around the lesion, we add a ruler along one border, 12–29 curved hairs, a dark vignette or 2–4 purple ink arcs to each test image. Conversely, detected thin dark structures (real hair and ruler marks, 544

TABLE III
ARTIFACT–MELANOMA ASSOCIATION (TRAINING SPLIT). OR: ODDS RATIO; MH: MANTEL–HAENSZEL OR STRATIFIED BY SOURCE.

Artifact	Prev.	$P(\mathrm{m}\|\mathrm{a})$	$P(\mathrm{m}\|\neg\mathrm{a})$	OR	MH OR [95% CI]
Dark corners	25.5%	.263	.149	2.03	1.555 [1.43, 1.69]
Bright structures	13.8%	.266	.164	1.85	1.751 [1.59, 1.92]
Thin dark struct.	24.2%	.174	.179	0.97	0.988 [0.90, 1.08]
Ink / patches	1.6%	.202	.178	1.18	1.114 [0.85, 1.46]
Any	48.8%	.210	.148	1.53	1.265 [1.17, 1.36]

images) are erased by Telea inpainting [35] outside a 2-pixel margin. As a null control, a same-shaped region on artifact-free skin is inpainted. We verify that every lesion pixel stays bit-identical and record the mean $|\Delta p|$ of the melanoma probability, the rate of flipped decisions and, for the three prototypes most active on the original image, the Spearman correlation ρ between the top-1 maps before and after the edit and the *capture*, the new hot region's share on inserted pixels.

Statistics. The 2,320 test images form 1,234 groups of the split (a lesion with its near-duplicates; 471 with several images), so we resample whole groups: ensemble AUCs get 95% CIs from 1,000 group resamples and are compared by a paired bootstrap with 2,000 (two-sided p), per-image metrics, averaged over seeds and then within groups, by the paired Wilcoxon test. Derm7pt (one image per case) uses the paired DeLong test [36], [37], PAD-UFES-20 a bootstrap over lesions. The primary family, corrected with Holm's method, holds the 16 main-claim tests: Ours against each of PPN-std, PPN, PPN-FA and the strongest mask-loss variant of Section VI-G on AUC, artifact share and flip rate under dark corners and rulers. All other tests are secondary and uncorrected. Image-level tests, which ignore the grouping, agree unless noted. These tests treat trained models as fixed and capture test-sample uncertainty; variation over training runs is shown by the std over seeds and tested over seven runs per model (exact Mann–Whitney). p-values below 10^{-6} are given as $p < 10^{-6}$.

VI. RESULTS

Main models: CNN, PPN-std, PPN, PPN-FA, Ours; ablations, mask-loss variants and $\mathcal{L}_{\mathrm{art}}$ follow in Sections VI-F–VI-H; bold: best of the four main prototype models.

A. Artifacts are associated with melanoma

In the training split, melanomas show dark corners and bright structures significantly more often than other lesions (Table III). Source could explain this (47% of BCN but 5% of HAM images have dark corners, and BCN has the highest melanoma rate), but the Mantel–Haenszel odds ratios stratified by source stay above 1: for dark corners in every source (BCN 1.41, HAM 3.10, MSK 2.24), for bright structures in BCN and HAM (2.11, 1.92) but not MSK (0.88). The strength differs between sources (Breslow–Day $p < 10^{-6}$), so pooled odds ratios are averages; thin dark structures and ink show no association.

B. Classification

On ISIC test, Ours has the highest AUC of the prototype models (0.865 ± 0.006, ensemble 0.884; Table IV). Its

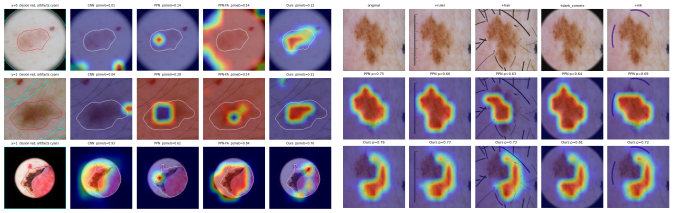


Fig. 2. (a, left) ISIC test images with artifacts (lesion red, detected artifacts cyan); CNN: class-activation map; prototype models: map of the most activated prototype; p: melanoma probability (seed 0). The CNN and PPN-FA respond to dark corners and borders, the top prototype of Ours stays on the lesion. (b, right) Counterfactual example with unchanged lesion pixels (not selected: the second of the first two test images that receive a ruler); rows 2–3: map of the prototype most active on the original image and p for PPN and Ours; the map of Ours keeps its place, that of PPN shifts with added hair.

ensemble exceeds those of PPN-std, PPN and PPN-FA by 0.022, 0.018 and 0.017 (95% CI $[0.009, 0.036]$, $[0.004, 0.032]$, $[0.005, 0.031]$; Holm-adjusted $p = 0.012$, 0.042, 0.029), and its seeds vary less (std 0.006 vs. 0.040 for PPN). Each of the three Ours seeds beats each of the 13 seeds of PPN-std, PPN (seven) and PPN-FA (worst 0.8592 vs. best 0.8526; Derm7pt-dermoscopic 0.7405 vs. 0.7343). One of seven Ours seeds (0.863 ± 0.012, range 0.842–0.880) falls below the best baseline seed; across runs, Ours exceeds these 13 seeds and the seven PPN seeds (0.821 ± 0.039; exact Mann–Whitney $p = 0.001$, 0.007). At the validation threshold, Ours trades sensitivity for specificity; at 90% sensitivity (test ROC), its specificity is highest among prototype models (64.8%; PPN-std, PPN, PPN-FA 58.9–60.2%, Correia 63.3%, CNN 69.6%; ensembles $p \geq 0.14$; over seven runs vs. PPN $p = 0.025$, vs. Correia $p = 0.90$), and a validation threshold for 90% sensitivity reaches 92.6% on test. The black-box CNN remains more accurate (0.912 vs. 0.884, $p = 0.009$), a price of about 3 AUC points for interpretability (4.6 for PPN). Without the last-layer re-optimization (five epochs, epoch chosen on validation), all prototype models score lower, but Ours stays best (0.846 ± 0.007; baselines 0.765–0.797, ablations 0.787–0.841), significantly so against the baselines ($p \leq 0.001$), and loses less (0.019 vs. 0.039–0.079).

Externally, Ours has the highest AUC of the main prototype models on Derm7pt-dermoscopic (0.749) and Derm7pt-clinical (0.657). On the dermoscopic images its ensemble beats PPN-FA and PPN-std ($p = 1.7 \times 10^{-5}$, 2.8×10^{-6}, also after Holm correction) but not PPN ($p = 0.20$), whose seeds vary strongly. On the clinical images, its advantage over PPN-std and PPN ($p = 0.023$, 0.046) does not survive Holm correction ($p = 0.069$, 0.092), and over PPN-FA is not significant ($p = 0.30$). On PAD-UFES-20 (52 melanoma images of 36 lesions), Ours beats PPN-std ($p = 0.001$) and does not differ significantly from PPN and PPN-FA ($p \geq 0.23$); its mean AUC there (0.647, std 0.108) is even below that of PPN (0.665). The black-box EPVT reports 0.837 on Derm7pt-dermoscopic [7] (melanoma vs. benign, other split, ViT-B/16); our CNN reaches 0.799.

C. Where the models look

The black-box CNN is the model most drawn to artifacts (Table V, Fig. 2(a)): in images with artifacts, 23.1% of its hot

TABLE IV

CLASSIFICATION (MEAN ± STD OVER 3 SEEDS; THRESHOLD FROM ISIC VALIDATION). ENS.: SEED-ENSEMBLE AUC WITH 95% CI; p: ENSEMBLE AGAINST OURS ON ISIC TEST (PAIRED BOOTSTRAP OVER LESION GROUPS). OURS: TWO CORE COMPONENTS AND THE AUXILIARY $\mathcal{L}_{\text{art}}$; OURS-NOART: THE TWO CORE COMPONENTS ALONE; PPN-FA: IAIA-BL LOSS ON 7×7 MAPS (ORIGINAL FORMULATIONS IN TABLE VIII).

	ISIC 2019 test (n = 2,320)						External AUC		
Model	AUC	BAcc	Sens.	Spec.	Ens. AUC [95% CI]	p	Derm7pt-derm	Derm7pt-clin	PAD-UFES-20
CNN (black box)	.902±.006	.814±.011	.808±.044	.820±.040	.912 [.884, .935]	.009	.799±.023	.704±.022	.721±.022
PPN-std	.835±.013	.752±.014	.769±.063	.734±.040	.861 [.827, .888]	.002	.677±.027	.624±.023	.587±.019
PPN	.825±.040	.758±.010	**.786±.068**	.730±.087	.866 [.830, .894]	.014	.688±.072	.618±.038	**.665±.016**
PPN-FA (7×7)	.844±.008	.744±.021	.689±.045	.800±.005	.866 [.830, .895]	.007	.696±.024	.640±.013	.630±.059
Ours	**.865±.006**	**.780±.014**	.734±.075	**.825±.046**	**.884 [.850, .910]**	–	**.749±.009**	**.657±.019**	.647±.108
Ours-noLes	.830±.015	.746±.030	.766±.091	.726±.032	.856 [.821, .885]	<.001	.717±.016	.636±.033	.628±.047
Ours-noArt	.859±.011	.772±.003	.733±.056	.811±.054	.877 [.845, .904]	.22	.752±.014	.637±.030	.678±.018
Ours-stdPush	.863±.005	.785±.002	.823±.043	.747±.047	.884 [.852, .911]	.86	.747±.005	.670±.019	.658±.108

TABLE V

WHERE THE MODELS LOOK (MEAN OVER 3 SEEDS). LES.: HOT-REGION SHARE ON THE LESION; ART.: SHARE ON ARTIFACTS IN IMAGES WITH ARTIFACTS; ENR.: ARTIFACT ENRICHMENT; PT.: MAXIMUM ON AN ARTIFACT; L-PR.: PROTOTYPES PROJECTED ONTO CELLS WITH ≥50% LESION (OF 20; 20 BY CONSTRUCTION UNDER THE CONSTRAINED PROJECTION, NOT RANKED); DC–INK: SHARE ON THE GIVEN ARTIFACT IN IMAGES THAT CONTAIN IT; D7: DERM7PT-DERMOSCOPIC.

	ISIC test					ISIC test, by artifact type				D7		
Model	Les.↑	Art.↓	Enr.↓	Pt.↓	L-pr.	DC	TS	SP	INK	Les.	Pt.	Enr.
CNN	.466	.231	1.52	.117	–	.365	.016	.017	.029	.605	.153	.63
PPN-std	.518	.082	0.54	.050	12.0	.123	**.010**	.015	.049	.749	.075	.25
PPN	.562	.066	0.46	.042	14.3	.098	**.010**	.012	.036	**.789**	.044	.24
PPN-FA	.530	.102	0.68	.050	10.0	.158	**.010**	**.010**	.056	.760	.077	.28
Ours	**.589**	**.048**	**0.37**	**.013**	20.0	**.067**	.011	.016	**.032**	.734	**.015**	**.18**
Ours-noLes	.565	.106	0.67	.053	20.0	.162	.011	.013	.044	.772	.057	.30
Ours-noArt	.594	.042	0.40	.018	20.0	.056	.012	.020	.040	.713	.017	.23
Ours-stdPush	.515	.086	0.58	.034	7.7	.129	.012	.019	.030	.633	.033	.35

region lies on them, 1.5 times their area share. The prototype models look at artifacts less than their area suggests, Ours least (4.8±1.5% of the hot region, maximum on an artifact in 1.3±0.5% of images; ±: std over seeds), with the largest lesion fraction (58.9%). In every Ours run, all 20 prototypes lie on lesion cells ($M^L \geq 0.5$; Fig. 3), whereas PPN-FA projects on average 10.0 onto lesion and 2.0 onto artifact cells. Most attention on artifacts falls on dark corners, where Ours has the smallest share (Table V). On the test sample, Ours differs significantly from each other main model in lesion share, artifact share, enrichment and pointing ($p \leq 5.6 \times 10^{-4}$; artifact shares in the primary family: Holm-adjusted $p \leq 2.2 \times 10^{-5}$). Across training runs, these values overlap with those of PPN and PPN-std (Section VI-D). On Derm7pt-dermoscopic, Ours again has the lowest artifact share, enrichment and pointing rate ($p \leq 8.6 \times 10^{-3}$), but the three prototype baselines place more of their hot region on the lesion ($p \leq 1.2 \times 10^{-4}$; Table V).

Receptive field of the prototypes. We measure the effective receptive field (ERF) of each prototype's source cell as the input-gradient magnitude of its latent vector on the source image (eight random sign projections). For every model, only 8–12% of its mass lies inside the cell, and half of it spreads over 8–11% of the image. For Ours, 69 ± 3% lies on the lesion (10th percentile over prototypes 42%), against 48–53% for PPN-std, PPN and PPN-FA (10th percentiles 11–21%), and 2.5% on detected artifacts (4.3%, 2.7%, 7.3%). This comes from where prototypes are taken, not from a narrower field: the constraint selects source images whose lesion covers 54% of the image (baselines 40%), and relative to the lesion area every model concentrates its ERF on the lesion by a median factor of only 1.2–1.3. Ours-stdPush drops to 47%, Ours-noLes and Ours-noArt keep 71–72%. For 13% of the prototypes of Ours (baselines 22–28%), the ERF draws less on the lesion than its area would, so anchoring covers the projected cell and most, not all, of its receptive field (Fig. 3).

D. Counterfactual robustness

Table VI summarizes the interventions (2,063–2,318 modified images per insertion type). Adding dark corners *increases* the melanoma probability of every model (mean signed change $+0.041$ CNN, $+0.048$ PPN, $+0.020$ Ours), matching the association in Table III; the models have absorbed the shortcut, Ours the least. Ours roughly halves the flip rate of PPN for dark corners (10.9%→5.2%) and rulers (7.5%→3.6%) and reduces it for ink (9.5%→5.6%) and hair (13.9%→9.2%); it also flips fewer decisions than the CNN for all four insertions. Against every other main model these reductions are significant for all four insertions ($p < 10^{-6}$ against the prototype baselines, also Holm-adjusted for dark corners and rulers; $p \leq 0.008$ against the CNN), as is the smaller $|\Delta p|$ against PPN ($p < 10^{-6}$) and, except for hair ($p = 0.09$), the CNN ($p \leq 2.3 \times 10^{-4}$). Removing real hair and ruler marks changes more decisions than the null control for every model (Ours 4.4% vs. 1.6%), so inpainting alone cannot explain this sensitivity.

Threshold and direction. At the validation threshold for 90% sensitivity, Ours still flips less than every baseline under dark corners (3.7% vs. 4.9–6.9%; $p \leq 0.004$), than the ProtoPNet baselines and the CNN under rulers (2.6% vs. 4.1–7.1%; $p \leq 0.002$) and than PPN-std, PPN, IAIA-BL and the CNN under ink ($p \leq 0.025$); under hair only than the CNN, and Correia flips fewer (9.8% vs. 11.4%, $p < 10^{-4}$). At the Youden threshold, dark corners and rulers mostly flip benign to melanoma (PPN 9.7 of 10.9%, 5.2 of 7.5%), hair the reverse (Ours 7.7%, PPN 7.8%). Ours has fewer benign-to-melanoma flips than all prototype baselines under each insertion (corners, ruler, hair, ink: 4.1, 2.2, 1.5, 1.8% vs. ≥4.8, 2.5, 3.2, 2.6%).

The explanations of Ours also move the least (Fig. 2(b)). Under added dark corners, its top-1 map keeps $\rho = 0.92$ (PPN

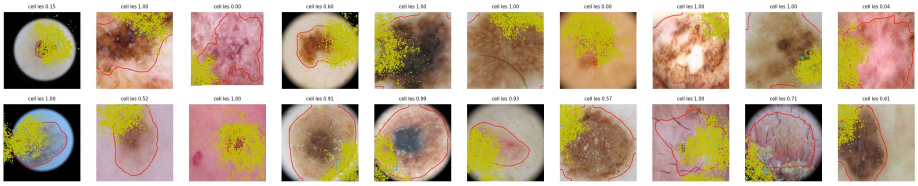


Fig. 3. Source images of the ten benign-class prototypes of PPN (top) and Ours (bottom), seed 0, with the projected cell (cyan; its lesion fraction above each image), the lesion (red) and the smallest region holding half of the prototype's effective receptive field (yellow). Four PPN prototypes come from background cells (lesion fraction 0.00–0.15) and draw on skin and vignette; all of Ours come from lesion cells (≥ 0.52, no detected artifact) and draw mostly, though not only, on the lesion.

TABLE VI

COUNTERFACTUAL INTERVENTIONS ON ISIC TEST (LESION PIXELS UNCHANGED; MEAN OVER 3 SEEDS). FLIP: DECISION CHANGES (%); $|\Delta p|$: MEAN ABSOLUTE CHANGE OF THE MELANOMA PROBABILITY; ρ: SPEARMAN CORRELATION OF THE TOP-1 PROTOTYPE MAP BEFORE/AFTER; CAP.: SHARE OF THE NEW HOT REGION ON INSERTED PIXELS (%). THE NULL CONTROL IS NOT RANKED.

	+Dark corners		+Ruler		+Hair		+Ink		Remove hair/ruler		Null control		Expl.: dark corners		Expl.: ruler	
Model	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	Flip	$\lvert\Delta p\rvert$	$\rho\uparrow$	Cap.$\downarrow$	$\rho\uparrow$	Cap.$\downarrow$
CNN	6.1	.049	5.6	.049	12.5	.104	9.7	.080	4.7	.036	1.0	.008	–	–	–	–
PPN-std	10.9	.060	8.3	.041	14.9	.092	9.9	.058	8.8	.036	4.3	.013	.635	7.1	.773	1.5
PPN	10.9	.067	7.5	.045	13.9	.096	9.5	.062	7.4	.045	3.6	.019	.680	6.1	.804	1.1
PPN-FA	10.6	.049	6.5	.028	15.1	.084	9.1	.046	7.2	**.030**	3.2	.010	.714	11.4	.874	1.1
Ours	**5.2**	**.035**	**3.6**	**.025**	**9.2**	**.081**	**5.6**	**.045**	**4.4**	.032	1.6	.011	**.918**	**3.4**	**.952**	**0.7**
Ours-noLes	10.2	.053	8.7	.040	14.3	.083	10.1	.049	8.3	.042	3.9	.016	.705	8.6	.827	1.7
Ours-noArt	5.4	.036	3.5	.023	8.7	.061	6.1	.041	4.4	.027	1.0	.011	.884	3.4	.949	1.0
Ours-stdPush	4.5	.034	4.1	.028	9.6	.077	5.8	.045	4.5	.033	1.9	.011	.923	7.5	.944	1.4

TABLE VII

REAL, UNEDITED ARTIFACTS ON BENIGN TEST LESIONS. A: PROBABILITY THAT A BENIGN LESION WITH THE ARTIFACT SCORES HIGHER THAN ONE WITHOUT IT (SEED ENSEMBLES; 0.5: NO EFFECT); FPR: FALSE-POSITIVE RATE (%) WITH/WITHOUT THE ARTIFACT (VALIDATION THRESHOLD, MEAN OVER SEEDS). BENIGN IMAGES WITH/WITHOUT: 447/1,483 (ISIC, DC), 251/1,679 (ISIC, SP), 515/244 (DERM7PT-CLIN., DC).

	ISIC, dark corners		ISIC, bright struct.		Derm7pt-clin., DC	
Model	$A\downarrow$	FPR	$A\downarrow$	FPR	$A\downarrow$	FPR
CNN	.57	22/17	.58	19/18	.59	18/9
PPN-std	.58	31/25	.61	32/26	.82	60/21
PPN	.61	30/26	.60	33/26	.78	35/11
PPN-FA	.66	33/16	.62	26/19	.80	46/15
Ours	**.57**	**20/17**	**.59**	**19/17**	**.69**	**24/12**
Ours-noLes	.66	37/25	.63	37/26	.77	50/24
Ours-noArt	.58	20/19	.62	25/18	.75	37/12
Ours-stdPush	.57	28/25	.59	29/25	.65	33/19

0.68), and only 3.4% of its hot region (per seed 0.2–7.7%) moves onto them (PPN 6.1%, PPN-FA 11.4%; $p < 10^{-6}$). Ink, drawn just outside the lesion border where anchored prototypes respond, is the exception: 1.8% of the hot region of Ours falls on it vs. 1.0% for PPN, though Ours still flips less.

Each of the seven Ours seeds flips fewer decisions under dark corners and ink than 12 of the 13 seeds of PPN-std, PPN and PPN-FA (3.3–7.5% vs. 7.6–15.4%; 4.4–8.0% vs. 8.0–11.2%, i.e., at most 184 vs. at least 185 of 2,301 images); the exception is the PPN seed with the lowest AUC (0.752; 2.0%, 6.4%). Across runs, Ours flips less than these 13 seeds under all four insertions ($p \leq 0.004$) and the seven PPN seeds ($p \leq 0.026$), not the seven Correia seeds ($p \geq 0.62$). Every Ours seed also keeps a more stable map than every seed of PPN-std and PPN-FA (ρ 0.85–0.93 vs. $\leq$0.77); ρ overlaps with two PPN seeds (0.87) and the mask-loss variants ($\leq$0.90). The artifact share (3.0–8.9%) lies below every CNN seed ($\geq$20.7%) but overlaps with the prototype baselines and Correia, and the captured share overlaps for all (Ours 0.2–7.7%, baselines 0.1–15.4%).

E. Shortcuts in unedited images

Without editing, we ask whether artifacts in the test images raise the melanoma score of *benign* lesions, measured by the threshold-free A of Table VII (0.5: no effect). On ISIC, dark corners double the false-positive rate of PPN-FA (16%→33%) but raise that of Ours only from 17% to 20%. The effect A of Ours (0.57) is significantly smaller than that of PPN, PPN-FA and Ours-noLes (0.61–0.66; $p < 0.001$) and does not differ significantly from that of PPN-std and the CNN ($p \geq 0.77$). Our weak mask loss thus leaves the real dark-corner shortcut in place, lesion attraction keeps it small, and the original mask-loss formulations also reduce it (Section VI-G). The same holds for bright structures, where Ours has the smallest effect of the prototype models (0.59 vs. 0.60–0.62; $p \leq 0.038$ against PPN-std and PPN-FA), and most clearly for dark border regions in clinical Derm7pt photographs (0.69 vs. 0.78–0.82; DeLong $p \leq 1.4 \times 10^{-6}$). The CNN attends to dark corners most, yet its effect is not larger than that of Ours.

F. Ablation of the two core components

Lesion attraction drives the accuracy gain. Without it (Ours-noLes), the test AUC falls to 0.830 ($p < 0.001$), the artifact share of the hot region more than doubles (4.8%→10.6%), and the flip rates for dark corners, rulers and ink roughly double. Ours-noLes still uses the constrained projection, so all its prototypes lie on lesion cells; a correct projection is not enough without anchoring the attention during training.

TABLE VIII

MASK-LOSS VARIANTS (ISIC TEST UNLESS NOTED; MEAN OVER 3 SEEDS). PPN-FA: 7×7 MAPS, BATCH MEAN; CORREIA [12], IAIA-BL [16]: 224×224 MAPS, BATCH SUM, $\lambda = 10^{-3}$. D7-D: DERM7PT-DERMOSCOPIC AUC; L-PR., ART.: AS IN TABLE V; FLIP, ρ: INSERTED DARK CORNERS; A: REAL DARK CORNERS; OURS+CORREIA: OURS PLUS THE CORREIA LOSS.

Mask loss	AUC	D7-d	L-pr.	Art.↓	Flip↓	$\rho\uparrow$	$A\downarrow$
None (PPN)	.825±.040	.688	14.3	.066	10.9	.68	.61
PPN-FA, 10^{-3}	.844±.008	.696	10.0	.102	10.6	.71	.66
PPN-FA, 10^{-2}	.833±.018	.695	16.0	.121	12.2	.80	.66
PPN-FA, 10^{-1}	.849±.006	.711	9.3	.061	7.2	.84	.60
Correia	.867±.011	**.771**	13.0	.141	6.5	.87	.57
IAIA-BL	.845±.011	.715	18.0	**.039**	6.2	.88	.58
Ours (no mask loss)	.865±.006	.749	20.0	.048	5.2	**.92**	.57
Ours+Correia	**.871±.004**	.736	20.0	.043	**5.0**	.87	**.53**

TABLE IX

ROBUSTNESS TO MASK QUALITY (ISIC TEST, MEAN OVER 3 SEEDS): ATTRIBUTION AS IN TABLE V ON IMAGES WITH EXPERT (HAM10000, $n = 900$) AND PREDICTED MASKS (BCN20000, MSK, $n = 1{,}420$); LESION SHARE WITH ALL LESION MASKS ERODED (-5) OR DILATED ($+5$) BY 5 PX; AUC PER SOURCE (n/MELANOMAS: BCN 1,151/248, HAM 900/93, MSK 269/49).

Model	Expert masks Les.↑	Art.↓	Pt.↓	Predicted masks Les.↑	Art.↓	Pt.↓	Lesion -5	$+5$	AUC by source BCN	HAM	MSK
CNN	.444	.043	.015	.480	.277	.182	.434	.496	.883	.934	.861
PPN-std	.416	.016	.004	.582	.098	.079	.480	.552	.825	.880	.711
PPN	.446	.014	.005	**.635**	.079	.065	.525	.595	.819	.860	.692
PPN-FA	.423	.019	.004	.599	.123	.080	.491	.567	.834	.896	.699
Ours	**.534**	**.013**	**.002**	.624	**.056**	**.021**	**.534**	**.640**	**.852**	**.906**	**.761**
Ours-noLes	.503	.023	.007	.603	.126	.082	.525	.601	.815	.879	.704
Ours-noArt	.543	.013	.003	.626	.050	.028	.541	.644	.853	.893	.737
Ours-stdPush	.459	.018	.003	.550	.103	.054	.466	.561	.852	.904	.749

The constrained projection determines what the prototypes *are* (Ours-stdPush: 7.7 of 20 on lesion cells). It does not change the accuracy (ensemble difference within $[-0.006, 0.005]$, 95% CI) or, significantly, the flip rates (dark corners 4.5% vs. 5.2%; $p \geq 0.13$), and the map differences of Ours-stdPush in Tables V and VI (artifact share 8.6% vs. 4.8%, captured share 7.5% vs. 3.4%) are not systematic. In a factorial control (Table X), we projected each network of seeds 3–6 of PPN, Correia and Ours both ways from its stored trained prototypes, refitting the last layer. In all 12 networks the constraint put the 20 prototypes on lesion cells (before 6–17, one Correia network 20; mean lesion fraction 0.62→0.90; exact Wilcoxon $p = 0.001$) and changed no measure of accuracy, attribution or robustness consistently (AUC $+0.0001$; others $p \geq 0.067$, both signs); seeds 0 and 2 of Ours and Ours-stdPush, which share their trained network (seed 1 diverged in training), agree. The training objective matters instead: with the same constrained projection, Ours keeps a higher ensemble AUC than PPN (0.886 vs. 0.866, $p = 0.001$), less artifact share (5.2% vs. 6.0%, $p < 10^{-5}$) and fewer flips under all four insertions (dark corners 6.0% vs. 10.5%; $p \leq 4 \times 10^{-4}$), as with the standard projection. Robustness thus comes from lesion attraction, not the projection.

We also varied the thresholds (seeds 3–6 of Ours). Lesion thresholds of 0.3–0.9 (artifact fraction ≤0.05) and artifact limits of 0, 0.2 or none (lesion fraction ≥0.5) put 14.8–20 of 20 prototypes on lesion cells (mean source-cell lesion fraction 0.78–0.99 vs. 0.47 unconstrained). The AUC (0.861–0.863), dark-corner flips (5.8–6.1%), artifact share (5.1–5.5%) and ρ (0.885–0.886) stay at the standard projection's (0.861, 5.8%, 5.2%, 0.886). Only the strictest (0.9) removes most border cells (2.3 vs. 8.5 at 0.5). The artifact limit itself excludes only 0.9% of training lesion cells (2 of 20,333 images lose all).

TABLE X

FACTORIAL CONTROL: SEEDS 3–6 OF EACH TRAINING OBJECTIVE, EACH NETWORK PROJECTED BOTH WAYS (ISIC TEST; MEAN OF 4 NETWORKS). L-PR., ART.: AS IN TABLE V; FLIP: DECISION CHANGES (%) UNDER INSERTED ARTIFACTS; ρ: DARK CORNERS.

Loss	Push	L-pr.	AUC	Art.↓	Flip↓ DC	Ruler	Hair	Ink	$\rho\uparrow$
None (PPN)	std.	11.8	.818	.062	10.4	6.9	15.3	9.2	.75
	constr.	20.0	.819	.060	10.5	6.1	15.1	8.0	.81
Correia	std.	16.0	.860	.042	6.1	3.6	10.5	6.9	.90
	constr.	20.0	.860	.042	6.1	3.6	10.4	6.8	.90
Ours	std.	9.0	.861	.052	5.8	3.8	10.4	6.7	.89
	constr.	20.0	.861	.052	6.0	4.1	10.7	6.8	.89

G. Strength of the mask-loss baseline

IAIA-BL [16] and Correia *et al.* [12] compute the fine-annotation loss at 224×224 and sum it over the batch; with the same $\lambda = 10^{-3}$, it is about three orders of magnitude stronger than in PPN-FA (factors 32 and 64; Correia *et al.* penalize own-class prototypes only). We trained both original formulations and PPN-FA with $\lambda = 10^{-2}$ and 10^{-1}, three seeds each (Table VIII; secondary). The ISIC AUC of the Correia formulation does not differ significantly from that of Ours (-0.001, 95% CI $[-0.012, 0.011]$), and it generalizes better (Derm7pt-dermoscopic 0.771 vs. 0.749, Derm7pt-clinical 0.702 vs. 0.657, PAD-UFES-20 0.699 vs. 0.647; $p \leq 0.028$), but only 13 of its 20 prototypes are lesion cells and its hot region lies three times more on artifacts (14.1% vs. 4.8%, $p < 10^{-6}$). In the primary family, Ours is significantly better on artifact share and dark-corner flips (Holm $p < 10^{-6}$, $p = 0.029$), not on AUC or ruler flips. Neither advantage replicates on seeds 3–6 (Table X): there, Correia puts 13–20 prototypes on lesion cells, has less artifact share (4.2% vs. 5.2%, $p < 0.001$) and flips as often ($p \geq 0.17$), while less of the hot region of Ours moves onto inserted dark corners (1.0–1.9% vs. 3.8–4.4%, $p < 10^{-5}$); it again generalizes better to Derm7pt-clinical (0.716 vs. 0.689, $p \leq 0.022$). Over seven seeds, its artifact share spans 0.5–24.6% (Ours 3.0–8.9%), and neither AUC nor flip rates differ across runs ($p \geq 0.62$). IAIA-BL keeps its hot region on the lesion (83%) and off artifacts (3.9%) better than Ours ($p < 0.002$) but is less accurate (0.845; ensembles $p = 0.057$) and leaves 2 of 20 prototypes off lesion cells. On seeds 0–2, every variant flips more often than Ours under dark corners (6.2–12.2% vs. 5.2%) and ink (6.6–10.7% vs. 5.6%; $p \leq 0.02$), though per seed the original formulations, and for dark corners $\lambda = 10^{-1}$, overlap with Ours; under rulers the original formulations ($p \geq 0.052$) and under hair Correia ($p = 0.37$) do not differ significantly, and none keeps its explanation as stable (Section VI-D). The real dark-corner effect of both original formulations does not differ significantly from that of Ours (A 0.57–0.58 vs. 0.57, $p \geq 0.26$), unlike our

weak version (0.66). Mask supervision thus shapes where the network looks and helps it generalize; anchoring matches its decision robustness, fixes what the prototypes are and keeps the artifact share consistent across runs; the two are complementary.

Adding the Correia loss to our model (Ours+Correia) gives the highest ISIC AUC of all prototype models (0.871 ± 0.004; $p \geq 0.32$ against Ours and Correia), maps more lesion-focused than those of Ours (lesion share 73% vs. 59%, seeds 59–85% vs. 49–69%; artifact share 4.3% vs. 4.8%, captured share 2.2% vs. 3.4%; $p < 10^{-6}$), the smallest real dark-corner effect of all models ($A = 0.53$; $p < 0.001$ against both) and similar flip rates ($p \geq 0.26$; hair 7.9% vs. 9.2%, $p = 0.02$). It does not inherit the Derm7pt advantage of the mask loss (0.736 and 0.675 vs. 0.771 and 0.702; DeLong $p \leq 0.016$) but has the highest PAD-UFES-20 AUC of all models (0.737, CNN 0.721; $p \leq 0.001$ against both), and its maps are less stable than those of Ours (ρ 0.87 vs. 0.92).

H. Artifact suppression: a largely negative result

The auxiliary term penalizes attention on artifacts directly. With $A_{ij} = \sum_{h,w} \alpha_{ij} M_i^A$, the attention mass of prototype j on the artifacts of image i,

$$\mathcal{L}_{\text{art}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{j=1}^{m} \omega_{ij} A_{ij}. \tag{3}$$

Removing it (Ours-noArt) changes the accuracy significantly neither on ISIC (ensemble difference 0.007, 95% CI $[-0.004, 0.018]$) nor on any external set ($p \geq 0.058$). Several counterfactual measures are even better without it (e.g., $|\Delta p|$ 0.061 vs. 0.081 for added hair, $p < 10^{-6}$), and the attribution changes little (artifact share 4.2% vs. 4.8%, $p = 0.36$; pointing 1.8% vs. 1.3%, $p = 5.7 \times 10^{-5}$). The explanation gains come from the two core components alone: against PPN-std, PPN and PPN-FA, Ours-noArt has more hot region on the lesion, less on artifacts and fewer flips under all four insertions ($p \leq 5.2 \times 10^{-4}$). Its AUC gain over them (0.010–0.016) is not significant with lesion groups resampled ($p = 0.073$–0.17; image-level DeLong $p = 0.005$ against PPN-std), so the significant accuracy gain belongs to the main model. $\mathcal{L}_{\text{art}}$ mainly weakens the response to real bright structures and to dark border regions in clinical images (A 0.59 vs. 0.62 and 0.69 vs. 0.75; $p < 0.001$, Table VII); otherwise its effect is small, so we keep it as an auxiliary term, not a contribution.

I. Robustness to mask quality

Predicted masks could bias the results if a model looked "on the lesion" only as judged by an imperfect mask, so we split the ISIC test set by mask provenance and perturb the lesion masks (Table IX). On the HAM10000 test images, which have expert masks, Ours places 53.4% of its hot region on the lesion (prototype baselines 41.6–44.6%, CNN 44.4%), a *larger* advantage than on the whole test set (+8.8 vs. +2.7 points over the best baseline), so it is not produced by errors of the predicted masks; its artifact scores are the lowest there too. On the predicted-mask images, Ours has the lowest artifact share (5.6% vs. 7.9–27.7%), although PPN places slightly more on the lesion (63.5 ± 1.5% vs. 62.4 ± 2.6%).

Eroding or dilating every lesion mask by 5 pixels shifts the lesion fractions of all models alike and keeps the order of the five main models (Ours first); artifact share, pointing and enrichment change by at most 0.011. The provenance guarantee itself depends on the mask border: with masks eroded by 5 pixels, on average 17 of the 20 prototypes of Ours remain on cells with $\geq$50% lesion (lowest 0.36), all 20 with dilated masks. Ours is also the most accurate of the four main prototype models in every source, with expert (HAM10000) and with predicted training masks (BCN20000, MSK; Table IX).

A predicted mask that is too small could let inserted artifacts touch the true lesion. On the expert-mask subset, where the protected region is exact, Ours flips the fewest decisions (dark corners 4.4% vs. 7.1% for the CNN and 10.0–10.3% for the prototype baselines; rulers 2.3% vs. 3.9% and 4.7–6.1%). On the predicted-mask subset it flips the fewest among the prototype models (dark corners 5.7% vs. 10.7–11.5%, CNN 5.5%; rulers 4.4% vs. 7.7–9.7%, CNN 6.7%), with the smallest $|\Delta p|$ in both. The conclusions thus hold on expert masks alone, under $\pm$5-pixel mask errors and in every source.

VII. Discussion and Limitations

The two components play different roles. Lesion attraction changes what the network learns, making it more accurate and less dependent on artifacts. The constrained projection changes what the network shows: it guarantees the provenance of each explanation, which matters for presentation and audit, but is not itself a mechanism against shortcuts. Our evidence thus supports claims about prototype provenance and robustness to artifacts, not superior accuracy: our AUC does not differ significantly from that of the strongest mask-supervised ProtoPNet, which is as robust across training runs, and stays below that of the black-box CNN. Lesion masks are needed only in training and come from public expert segmentations of one source, so a new dermoscopic dataset needs no new annotation; as coarse 7×7 cell fractions, they tolerate boundary errors (Section VI-I). Anchoring retains the lesion border: 9.3 of 20 prototypes of Ours are border cells (lesion fraction 0.5–0.95; baselines and mask-loss variants 1.7–5.0), so ink just outside the border draws some of its attention (Section VI-D).

Limitations. (i) A gap of about 3 AUC points to the black-box CNN remains; Correia *et al.* [12] found prototype models on par with CNNs, but on ISIC validation data and after choosing the best of many configurations. (ii) Predicted BCN20000 and MSK masks are validated only indirectly (held-out HAM10000 images, PH2, subset analysis), and the artifact detectors miss many artifacts (Section IV), so our evidence concerns detected and inserted artifacts only. (iii) Inserted artifacts are synthetic and may lie outside the training distribution. Removing real artifacts, a null control and the test on unedited images mitigate this, but on unedited images Ours is significantly better only than PPN, PPN-FA and Ours-noLes, possibly partly due to lesion appearance. (iv) The 7×7 grid limits the spatial resolution of the explanations, and

class-activation and prototype maps differ in kind, so attribution comparisons between the CNN and the prototype models are only indicative. (v) External gains over the prototype baselines are only partly significant (Section VI-B), and the original mask loss generalizes better (Section VI-G). (vi) We studied one backbone and one prototype architecture, as a controlled comparison requires, although both components apply to any prototype network with a projection step. Training variability rests on three seeds per configuration (seven for PPN, Correia and Ours; 12 paired networks for the projection), between which explanation metrics vary strongly (Section VI-D). (vii) No dermatologist rated the prototypes, and clinical readiness would need prospective validation in other populations.

VIII. Conclusion

Classifiers trained on ISIC 2019 exploit artifact shortcuts. We anchored a prototype network to the lesion without new annotation, by lesion attraction on the prototype attention and a projection restricted to lesion-dominated cells. In a same-backbone, leakage-controlled comparison, lesion attraction made ProtoPNet more accurate and reduced its decision flips under all four inserted artifacts, across seven training runs and, for dark corners and rulers, also at 90% sensitivity; ProtoPNet with the projection alone stayed less accurate and robust. The projection fixed provenance without changing either in 12 networks, and the auxiliary artifact-suppression term had small, mixed effects. The strongest mask-supervised ProtoPNet was as accurate and, across runs, as robust and generalized better to Derm7pt, but kept all prototypes on lesion-dominated cells in one of seven runs; adding it to our model further weakened the real dark-corner shortcut. **Future work** will test lesion anchoring in other prototype architectures (e.g., PIP-Net [13]) and backbones, seek a combination with mask supervision that keeps its external generalization, use learned, higher-recall artifact detectors and evaluate the prototypes with dermatologists in a reader study. The same counterfactual protocol can check other interpretable classifiers.

References

[1] A. Esteva *et al.*, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.

[2] R. Geirhos *et al.*, "Shortcut learning in deep neural networks," *Nat. Mach. Intell.*, vol. 2, pp. 665–673, 2020.

[3] J. K. Winkler *et al.*, "Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition," *JAMA Dermatol.*, vol. 155, no. 10, pp. 1135–1141, 2019.

[4] A. Bissoto *et al.*, "(De)constructing bias on skin lesion datasets," in *Proc. IEEE/CVF CVPR Workshops*, 2019.

[5] A. Bissoto *et al.*, "Debiasing skin lesion datasets and models? Not so fast," in *Proc. IEEE/CVF CVPR Workshops*, 2020.

[6] A. Bissoto *et al.*, "Artifact-based domain generalization of skin lesion models," in *Proc. ECCV Workshops*, 2022.

[7] S. Yan *et al.*, "EPVT: Environment-aware prompt vision transformer for domain generalization in skin lesion recognition," in *Proc. MICCAI*, 2023.

[8] S. W. Pewton and M. H. Yap, "Dark corner on skin lesion image dataset: Does it matter?" in *Proc. IEEE/CVF CVPR Workshops*, 2022, pp. 4831–4839.

[9] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nat. Mach. Intell.*, vol. 1, pp. 206–215, 2019.

[10] C. Chen *et al.*, "This looks like that: Deep learning for interpretable image recognition," in *Proc. NeurIPS*, vol. 32, 2019.

[11] A. Hoffmann *et al.*, "This looks like that... does it? Shortcomings of latent space prototype interpretability in deep networks," in *Proc. ICML Workshop on XAI*, 2021.

[12] M. Correia *et al.*, "XAI for skin cancer detection with prototypes and non-expert supervision," in *Proc. MICCAI Workshop iMIMIC*, 2023, arXiv:2402.01410.

[13] M. Nauta *et al.*, "Interpreting and correcting medical image classification with PIP-Net," in *Proc. ECAI Workshops*, CCIS vol. 1947, 2024, pp. 198–215.

[14] P. Bevan and A. Atapour-Abarghouei, "Skin deep unlearning: Artefact and instrument debiasing in the context of melanoma classification," in *Proc. ICML*, 2022.

[15] F. Pahde *et al.*, "Reveal to revise: An explainable AI life cycle for iterative bias correction of deep models," in *Proc. MICCAI*, 2023.

[16] A. J. Barnett *et al.*, "A case-based interpretable deep learning model for classification of mass lesions in digital mammography," *Nat. Mach. Intell.*, vol. 3, pp. 1061–1070, 2021.

[17] A. Bontempelli *et al.*, "Concept-level debugging of part-prototype networks," in *Proc. ICLR*, 2023.

[18] M. Nauta *et al.*, "Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis," *Diagnostics*, vol. 12, no. 1, art. 40, 2022.

[19] X. Lin and M. Wright, "Diverse by design: Architectural constraints for prototype-based interpretability," arXiv:2609.27194, 2026.

[20] J. Adebayo *et al.*, "Sanity checks for saliency maps," in *Proc. NeurIPS*, vol. 31, 2018.

[21] A. S. Ross *et al.*, "Right for the right reasons: Training differentiable models by constraining their explanations," in *Proc. IJCAI*, 2017, pp. 2662–2670.

[22] L. Rieger *et al.*, "Interpretations are useful: Penalizing explanations to align neural networks with prior knowledge," in *Proc. ICML*, 2020, pp. 8116–8126.

[23] G. Huang *et al.*, "Densely connected convolutional networks," in *Proc. IEEE CVPR*, 2017, pp. 4700–4708.

[24] P. Tschandl *et al.*, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, art. 180161, 2018.

[25] M. Combalia *et al.*, "BCN20000: Dermoscopic lesions in the wild," arXiv:1908.02288, 2019.

[26] N. C. F. Codella *et al.*, "Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC)," in *Proc. IEEE ISBI*, 2018, pp. 168–172.

[27] L.-C. Chen *et al.*, "Rethinking atrous convolution for semantic image segmentation," arXiv:1706.05587, 2017.

[28] A. Howard *et al.*, "Searching for MobileNetV3," in *Proc. IEEE/CVF ICCV*, 2019, pp. 1314–1324.

[29] T. Mendonça *et al.*, "PH^2 – A dermoscopic image database for research and benchmarking," in *Proc. IEEE EMBC*, 2013, pp. 5437–5440.

[30] T. Lee *et al.*, "DullRazor: A software approach to hair removal from images," *Comput. Biol. Med.*, vol. 27, no. 6, pp. 533–543, 1997.

[31] J. Kawahara *et al.*, "Seven-point checklist and skin lesion classification using multitask multimodal neural nets," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 2, pp. 538–546, 2019.

[32] A. G. C. Pacheco *et al.*, "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," *Data in Brief*, vol. 32, art. 106221, 2020.

[33] R. R. Selvaraju *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE ICCV*, 2017, pp. 618–626.

[34] B. Zhou *et al.*, "Learning deep features for discriminative localization," in *Proc. IEEE CVPR*, 2016, pp. 2921–2929.

[35] A. Telea, "An image inpainting technique based on the fast marching method," *J. Graphics Tools*, vol. 9, no. 1, pp. 23–34, 2004.

[36] E. R. DeLong *et al.*, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[37] X. Sun and W. Xu, "Fast implementation of DeLong's algorithm for comparing the areas under correlated receiver operating characteristic curves," *IEEE Signal Process. Lett.*, vol. 21, no. 11, pp. 1389–1393, 2014.